

Temporal Analysis of NDVI Trends and Prediction of Green Area Coverage Using LSTMs

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I. Introduction

The normalized difference vegetation index (NDVI) is a widely-used metric for quantifying the health and density of vegetation using sensor data. It is calculated from spectrometric data at two specific bands: red and near-infrared. The spectrometric data is usually sourced from remote sensors, such as satellites.

NDVI (Normalized Difference Vegetation Index) monitors vegetation health and coverage by comparing near-infrared and red light reflectance from satellite or aerial imagery. It helps distinguish healthy vegetation from stressed or sparse areas, aiding in agriculture, environmental management, urban planning, and disaster recovery.

QGIS is used for analyzing geospatial data. It measures greenery using indices like NDVI and classifies vegetation types. It crops images with tools like Clip and Warp, and plugins such as SCP enhance functionality for remote sensing tasks.

When working with remote sensing data such as NDVI (Normalized Difference Vegetation Index) values, Colab enables calculations and visualizations to assess greenery levels over time or across different regions. Its seamless integration of coding, data handling, and visualization makes it a powerful tool for researchers, analysts, and developers alike.

II. Abstract

The Normalized Difference Vegetation Index (NDVI) is a widely used metric for quantifying the health and density of vegetation using sensor data. Calculated from spectrometric data at red and near-infrared bands, NDVI is typically sourced from remote sensors like satellites, providing critical insights into vegetation health and coverage. This metric plays a pivotal role in various fields, including agriculture, environmental management, urban planning, and disaster recovery, by distinguishing healthy vegetation from stressed or sparse areas.

In this research, we employ geospatial analysis tools such as QGIS to measure and classify vegetation types using NDVI and other indices. QGIS facilitates image cropping through tools like Clip and Warp, while plugins such as SCP enhance remote sensing capabilities. For advanced data processing and visualization, we utilize Google Colab, which enables seamless integration of coding, data handling, and visualization, thus allowing for comprehensive assessments of greenery levels over time or across different regions.

Furthermore, we integrate Long Short-Term Memory (LSTM) models to predict NDVI trends, addressing the limitations of traditional statistical models in capturing temporal dependencies in sequential data. This study not only enhances the accuracy of NDVI predictions for a specific area but also establishes a framework applicable to other geographical regions. By combining advanced geospatial analysis techniques and machine learning, this research contributes significantly to the fields of geospatial analysis and environmental monitoring.

III. Problem Statement

In recent years, the analysis of geospatial data using NDVI (Normalized Difference Vegetation Index) has become increasingly crucial for monitoring environmental changes and agricultural productivity. However, existing methods often face challenges in accurately predicting NDVI trends over time, particularly when using traditional statistical models. This research addresses these limitations by employing Long Short-Term Memory (LSTM) models, known for their capability to capture temporal dependencies in sequential data, to predict NDVI values over a specific area. The study aims to not only enhance the accuracy of NDVI predictions but also to establish a framework that can be applied to other geographical regions beyond the specific area studied, thereby contributing to the broader field of geospatial analysis and environmental monitoring.

IV. Literature Survey

1.	Title:	A Machine Learning Approach for NDVI Forecasting Based on Sentinel-2 Data
	Authors:	Stefano Cavalli, Gabriele Penzotti, Michele Amoretti and Stefano Caselli at CIDEA, University of Parma, Parco Area delle Scienze, Parma, Italy
	Summary:	The paper presents a machine-learning approach for forecasting the Normalized Difference Vegetation Index (NDVI) using Sentinel-2 satellite data. The goal is to predict NDVI values based on historical data, employing machine learning techniques to enhance the accuracy of forecasting. The use of Sentinel-2 data in the paper provides guidance on handling and processing satellite imagery, which is crucial for our own model development. Additionally, the temporal analysis discussed in the paper aligns with our objective of examining NDVI trends over an extended period. Learning from the machine learning techniques, methodological insights, and preprocessing steps in the paper will aid in the accurate estimation of greenery on the campus using historical NDVI data.
2.	Title:	GDAL and PROJ Libraries Integrated with GRASS GIS for Terrain Modelling of the Georeferenced Raster Image
	Authors:	Polina Lemenkova and Olivier Debeir (École Polytechnique de Bruxelles, Belgium)
	Summary:	This paper demonstrates the use of GDAL and PROJ libraries for manipulating and converting raster data, similar to satellite images used in NDVI analysis. They use GRASS GIS modules for visualizing and analyzing topographic data, including slope, aspect, profile, tangential curvature, depth frequency, and contour. The integration of these tools with GRASS GIS for detailed topographic visualization can inform our approach in visualizing NDVI data.

3.	Title:	Okinawa Trough Geophysical and Topographic Modeling by GDAL Utilities and GRASS GIS
	Authors:	Polina Lemenkova
	Summary:	The author discusses the use of GDAL (Geospatial Data Abstraction Library) and GRASS GIS (Geographic Resources Analysis Support System) in collaboration with GDAL library to manipulate raster data using GDAL utilities such as gdaldem, gdalwarp, gdalinfo, and gdal_translate for data processing, as well as shedding light on the use of GRASS GIS modules for data visualization. The study focuses on the Okinawa Trough, Ryukyu trench-arc system, southern Japan, East China Sea, and the Philippine Sea. The study demonstrates the generation of topographic models, including aspect, hillshade, roughness, and topographic indices. It also visualizes thematic maps like topography, geoid, and marine free-air gravity. The results show variations in bathymetry and relief patterns in different areas, corresponding to their geological and geophysical settings. The methodology presented in the paper highlights the technical aspects of using GDAL and GRASS GIS for advanced cartography.
4.	Title:	Evolution and use of remote sensing in ecological vulnerability assessment: A review
	Authors:	Muhammad Kamran, Kayoko Yamamoto, Japan
	Summary:	The text discusses the role of Remote Sensing (RS) in conducting Ecological Vulnerability Assessment (EVA) research. This paper maps the detailed review of 80 peer-reviewed journal articles that use RS technology for EVA. It covers various aspects of EVA research, including the general trend and characteristics, spatiotemporal scale of studies, target systems investigated, factors influencing the inclusion or exclusion of indicators, nature and number of indicators, types of data sources used, satellites and sensors employed, conceptual models and weighing mechanisms used in the literature. The most commonly used GIS is ARCGIS, which is what we'll be working with, hence making this paper a not insignificant resource.

5.	Title:	A Machine Learning Approach for Mapping Forest Vegetation in Riparian Zones in an Atlantic Biome Environment Using Sentinel-2 Imagery
	Authors:	Danielle Elis Garcia Furuya, João Alex Floriano Aguiar, Nayara V. Estrabis, Mayara Maezano Faita Pinheiro, Michelle Taís Garcia Furuya, Danillo Roberto Pereira, Wesley Nunes Gonçalves , Veraldo Liesenberg, Jonathan Li, José Marcato Junior, Lucas Prado Osco and Ana Paula Marques Ramos
	Summary:	The research in this paper proposes a machine learning method to map forest vegetation in riparian zones within an Atlantic biome using Sentinel-2 satellite imagery. The study leverages advanced techniques to analyze the satellite data and accurately identify and map vegetation in these specific areas. This approach can enhance our understanding of the forest ecosystem in riparian zones and contribute to effective environmental management in the Atlantic biome environment.
6.	Title:	A GIS-based assessment of green space accessibility: case study of Dundee
	Authors:	Assel Kmail and Vincent Onyango
	Summary:	The paper aimed to evaluate the accessibility of green spaces in Dundee City using GIS, focusing on the potential health benefits. The Accessibility to Natural Green Spaces Tool (ANGSt) was chosen as the benchmark due to its proven success in previous studies. The research highlighted that access to green spaces offers numerous health benefits, such as promoting physical activity, reducing stress, accelerating patient recovery, and preventing violence. The study involved collecting GIS datasets from various sources for network analysis. The findings revealed that, according to ANGSt standards, certain areas in Dundee have good access to green spaces.

7.	Title:	Satellite Image Processing by Python and R Using Landsat 9 OLI/TIRS and SRTM DEM Data
	Authors:	Jerome Gilles
	Summary:	The paper emphasizes the importance of terrain analysis in image processing for Earth modeling and understanding environmental dynamics. Using Python and R, the study presents an integrated approach to processing satellite images, showcasing automation capabilities and the efficiency of high-level programming languages. Advanced algorithms for topographic and environmental modeling demonstrate the benefits of Python's machine-based automation and R's spatial data processing. The successful mapping of diverse Earth features underscores the need for advanced methods to analyze complex spatial patterns. Future work will involve higher-resolution images and long-term environmental dynamics using time series data. The study highlights the utility of map algebra and calculation techniques with Landsat OLI/TIRS for vegetation health monitoring and environmental mapping, leveraging R's "raster" and "terra" packages for computing vegetation indices. Additionally, a Python-based method using the EarthPy library is proposed to extract topographic data from the SRTM DEM for terrain modeling. This approach is also applicable to agricultural settings requiring close observation or remediation.
8.	Title:	A Sustainable Evaluation Study Based on Energy–Geographic Information System (GIS) Methodology in Hangzhou City from 2010 to 2035
	Authors:	Lan Liu, Runhui Cai and Junxue Zhang
	Summary:	This study utilizes the energy–GIS method to evaluate urban sustainability in Hangzhou city from 2000 to 2035. The integration of energy data with GIS effectively validates the city's sustainability status, providing a comprehensive methodology for future assessments. Challenges include differences between energy and GIS data, inconsistent unit energy values, and lack of detailed urban data, which can affect results, especially in larger cities. To address these issues, the study emphasizes the need for comprehensive urban data and consistent energy values. It suggests enhancing the energy–GIS method by incorporating multi-criteria decision analysis (MCDA) and life cycle assessment (LCA) for a more holistic sustainability evaluation, considering environmental, economic, and social factors. Future research should focus on overcoming data challenges and improving integration methods for better urban sustainability insights. The comprehensive approach enables policymakers to develop strategies that balance environmental, social, and economic sustainability, promoting sustainable development in cities.

9.	Title:	Application of remote sensing and machine learning algorithms for forest fire mapping in a Mediterranean area
	Authors:	Meriamé Mohajane a b 1, Romulus Costache c 1, Firoozeh Karimi d, Quoc Bao Pham e, Ali Essahlaoui b, Hoang Nguyen f g, Giovanni Laneve h, Fatiha Oudija a
	Summary:	This study develops and compares five hybrid machine learning algorithms—FR-MLP, FR-LR, FR-CART, FR-SVM, and FR-RF—for mapping forest fire susceptibility in northern Morocco. Using a dataset of 510 historical fire points and ten causal factors such as elevation, slope, and NDVI, the models are evaluated via the area under the receiver operating characteristics (ROC) curves (AUC). The RF-FR model performs best with an AUC of 0.989, followed by SVM-FR (0.959), MLP-FR (0.858), CART-FR (0.847), and LR-FR (0.809). These results, presented as prediction maps, support forest management by identifying high-risk areas, showcasing the potential of these models for broader application in regions facing similar challenges.
10.	Title:	NDVI Forecasting Model Based on the Combination of Time Series Decomposition and CNN-LSTM
	Authors:	Peiqiang Gao, Wenfeng Du E, Qingwen Lei, Juezhi Li, Shuaiji Zhang & Ning Li
	Summary:	This study presents a novel NDVI forecasting model combining time series decomposition (TSD), convolutional neural networks (CNN), and long short-term memory (LSTM). The model's performance was validated using two climatic factor models and four NDVI forecasting models. The TSD-CNN-LSTM model achieved superior prediction accuracy, with an RMSE of 0.0573, NSE of 0.9617, and MAE of 0.0447. The TP-N (Temperature & Precipitation-NDVI) model demonstrated a more significant impact compared to the T-N (Temperature-NDVI) and P-N (Precipitation-NDVI) models. The study concludes that NDVI changes are primarily driven by temperature and precipitation, with temperature being the most significant factor. These results offer valuable insights for understanding vegetation growth in response to climate changes.

V. Data acquisition

We started our experiment by gathering high-resolution Landsat satellite images with a particular focus on the Near-Infrared (NIR) and Red spectral bands from the USGS Earth Explorer. These bands were selected because they are effective for vegetation analysis due to their sensitivity to chlorophyll content in plants.

After acquiring the required imagery, we carefully cropped the photos to our region of interest, the K.J. Somaiya College of Engineering campus, using QGIS, a potent open-source geographic information system. We produced a detailed campus layout map with QGIS, ensuring precise borders and accurate representation for our research area.

We then examined the changes in greenery over a significant time span from 1988 to 2021 using the cropped NIR and Red images. This extensive investigation allowed us to track and measure the changes in the amount of vegetation on campus. By computing indices like the Normalized Difference Vegetation Index (NDVI), we gained important insights into trends and changes in the environment over the previous three decades.

Our statistical procedures and analyses produced quantifiable data, which we recorded and organized in Microsoft Excel spreadsheets. This made it easier to organize the data effectively and provided the opportunity to display our conclusions better with the help of graphs and charts. Utilizing Excel's powerful data analysis features, we identified patterns in the dynamics of the campus's vegetation and derived significant conclusions.

VI. Implementation

I. Data Preprocessing:

Our project began by acquiring high-resolution Landsat satellite images, specifically focusing on the Near-Infrared (NIR) and Red spectral bands from the USGS Earth Explorer. These bands are critical for vegetation analysis due to their sensitivity to chlorophyll content in plants. Using QGIS, we extracted the NIR and Red band data for our region of interest, the K.J. Somaiya College of Engineering campus. This process resulted in a series of 2D arrays or "maps", each containing the spectral band values for the specified area. The regions outside the area of interest were masked to zero.

Next, we calculated the Normalized Difference Vegetation Index (NDVI) for each map using the formula:

$$NDVI = \frac{(NIR - R)}{(NIR + R)}$$

This step was repeated for all the years, generating a dataset of NDVI maps over this period.

For each NDVI map, we conducted statistical analyses to identify trends and patterns. Key statistical values such as quartiles, mean, and median NDVI values were computed to provide insights into the distribution and central tendencies of vegetation health over time. Additionally, we introduced a novel metric called the Fixed Coordinate Limit (FCL). The FCL metric identifies a specific coordinate that remains green throughout the analysis period and calculates the percentage of the area with NDVI values greater than or equal to the NDVI value at this fixed coordinate. This metric serves as a robust indicator of consistent greenery within the area.

The formula for FCL is:

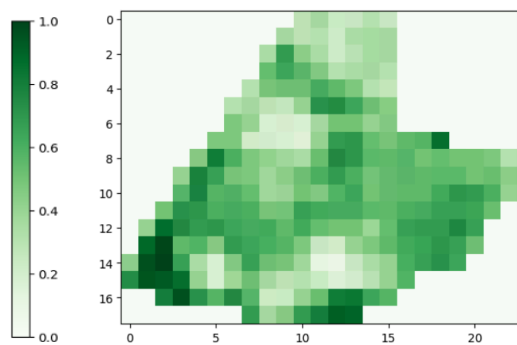
$$FCL = \left(\frac{\text{Number of pixels with NDVI} \geq \text{Fixed Point NDVI}}{\text{Total Number of Pixels}} \right) \times 100$$

The trends in these statistical values and the FCL greenery percentage were analyzed to observe changes in the campus's vegetation health over the years. Our analysis aimed to detect any downward trends in greenery, indicating potential environmental changes or degradation.

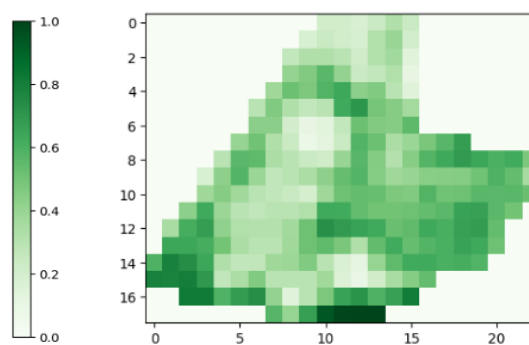
II. Data acquisition table after preprocessing:

Year	Date	Landsat version	Min value	Quartile 1	Quartile 2	Quartile 3	Max value	Mean	Median	Green Area coverage (%)
1989	Nov 1989	L-5	0	0	0.0960465	0.193954	0.337733	0.102738	0.0960465	39.13043
1992	Nov 1992	L-5	0	0	0.0767918	0.136912	0.250788	0.0747674	0.0767918	33.33333333
1993	Nov 1993	L-5	0	0	0.0864006	0.152091	0.312333	0.0843875	0.0864006	22.9468599
1994	Dec 1994	L-5	0	0	0.0685174	0.10494	0.188097	0.0589397	0.0685174	10.38647343
1995	Dec 1995	L-5	0	0	0.0722648	0.137326	0.269856	0.0736547	0.0722648	24.87922705
1996	Dec 1996	L-5	0	0	0.09124	0.157677	0.261568	0.0862105	0.09124	26.8115942
1997	Nov 1997	L-5	0	0	0.090101	0.153561	0.278189	0.0844542	0.090101	8.45410628
1998	Nov 1998	L-5	0	0	0.10641	0.182274	0.320079	0.101506	0.10641	15.70048309
1999	Nov 1999	L-5	0	0	0.0962012	0.184225	0.328506	0.100249	0.0962012	12.80193237
2000	Oct 2000	L-7	0	0	0.0890936	0.147753	0.298658	0.0827233	0.0890936	14.73429952
2001	Oct 2001	L-7	0	0	0.119456	0.194521	0.295824	0.106531	0.119456	9.178743961
2002	Oct 2002	L-7	0	0	0.110952	0.180082	0.294546	0.100331	0.110952	16.90821256
2003	Oct 2003	L-7	0	0	0.125266	0.221189	0.378018	0.12123	0.125266	20.04830918
2004	Nov 2004	L-7	0	0	0.122323	0.193992	0.333613	0.106358	0.122323	12.07729469
2005	April 2005	L-7	0	0	0.0749216	0.11158	0.287862	0.0647605	0.0749216	25.36231884
2008	Oct 2008	L4-5	0	0	0.0985166	0.169856	0.298656	0.0960278	0.0985166	11.352657
2009	Oct 2009	L4-5	0	0	0.111774	0.185605	0.31242	0.104912	0.111774	11.5942029

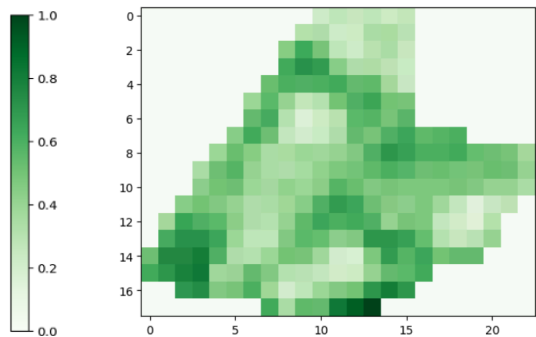
2010	April 2010	L-5	0	0	0.110644	0.18533	0.285821	0.100898	0.110644	5.314009662
2011	Feb 2011	L-5	0	0	0.0779437	0.126713	0.22243	0.0697	0.0779437	8.937198068
2012	Nov 2012	L-7	0	0	0.102235	0.14007	0.223516	0.0812506	0.102235	7.729468599
2014	Oct 2014	L8-9	0	0	0.12576	0.187913	0.293378	0.104048	0.12576	14.25120773
2015	Dec 2015	L8-9	0	0	0.0946209	0.138656	0.236504	0.0793195	0.0946209	2.657004831
2016	Dec 2016	L8-9	0	0	0.0862621	0.12611	0.199071	0.0718512	0.0862621	3.623188406
2017	Nov 2017	L8-9	0	0	0.101492	0.15157	0.245029	0.0862416	0.101492	2.898550725
2018	Nov 2018	L8-9	0	0	0.11822	0.188193	0.292087	0.10496	0.11822	2.415458937
2019	Nov 2019	L8-9	0	0	0.133583	0.209558	0.340697	0.117366	0.133583	2.657004831
2020	Nov 2020	L8-9	0	0	0.112408	0.170779	0.30239	0.095578	0.112408	2.898550725
2021	Nov 2021	L8-9	0	0	0.115136	0.181791	0.288424	0.101573	0.115136	2.173913043
2022	Nov 2022	L8-9	0	0	0.099896	0.155641	0.252128	0.0872412	0.099896	1.207729469
2023	Nov 2023	L8-9	0	0	0.11606	0.165701	0.266603	0.0952104	0.11606	1.207729469
2024	April 2024	L8-9	0	0	0.0976723	0.137252	0.233295	0.0801818	0.0976723	0.9661835749



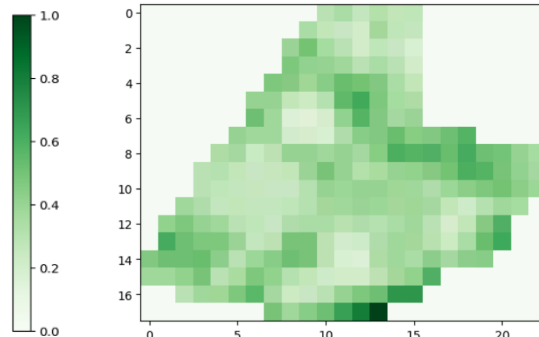
a. Nov 1989



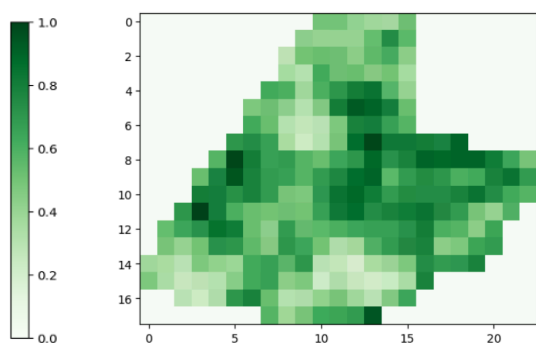
b. Dec 1995



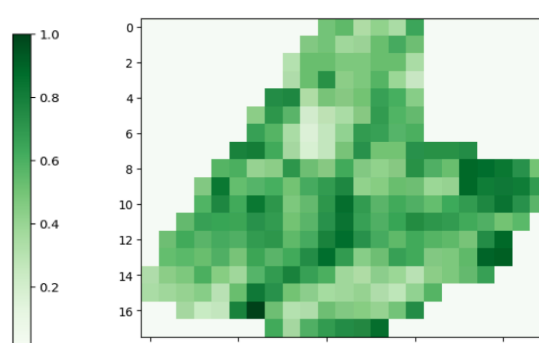
c. Oct 2000



d. April 2005



e. April 2010



f. Dec 2015

Fig VI.I.1 NDVI Map (a.Nov 1989, b.Dec 1995, c. Oct 2000, d. April 2005, e. April 2010, f. Dec 2015)

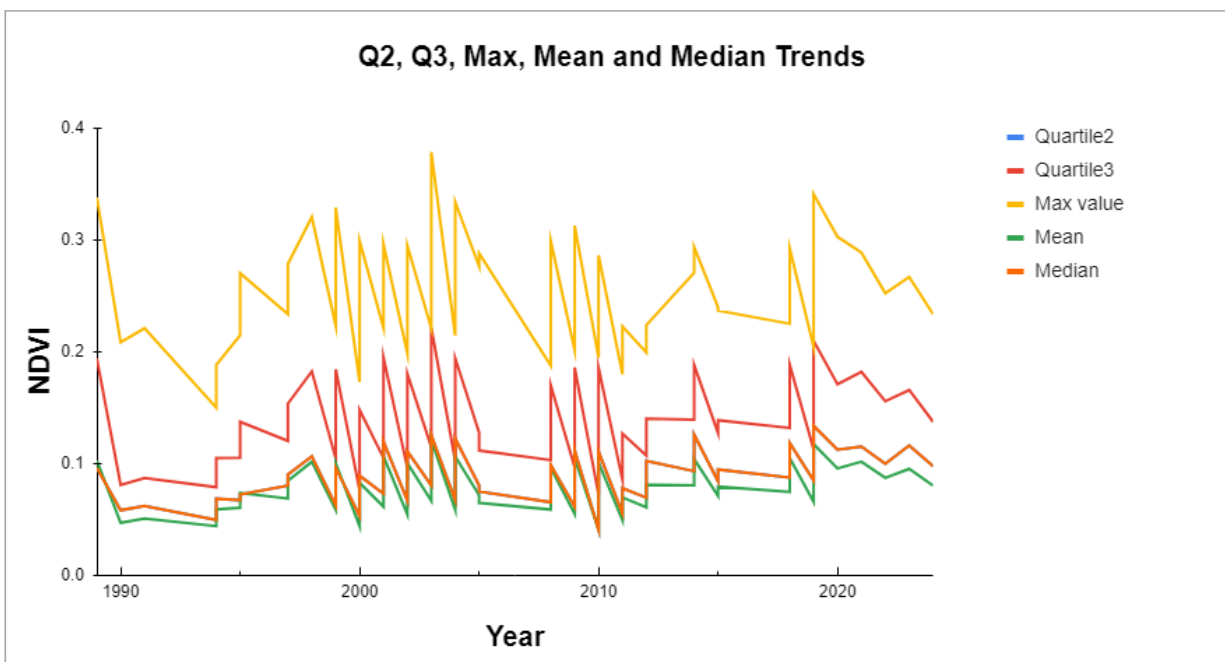


Fig VI.I.2 Q2, Q3, Max, Mean and Median NDVI Trend over the years

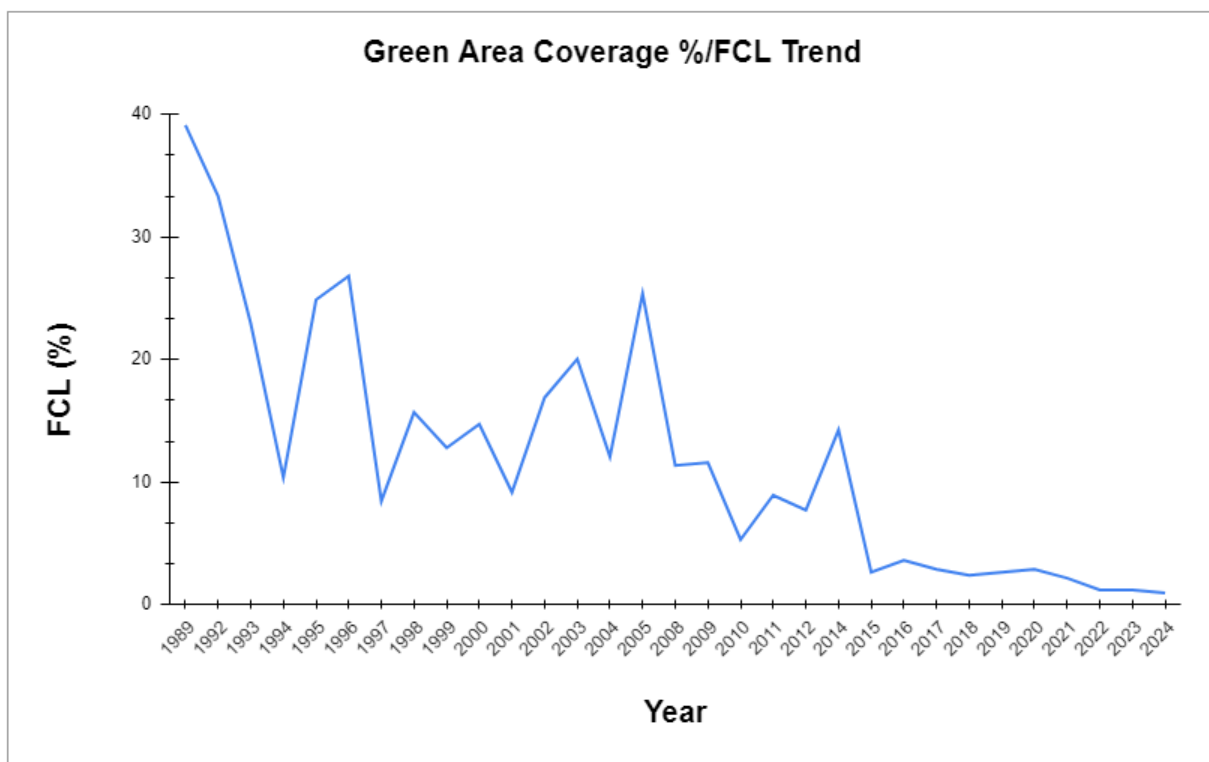


Fig VI.I.3 FCL Trend over the years

VII. Algorithms

A simple Linear Regression model or models only using the historical FCL values as input are not able to accurately capture the correct trend and nature of FCL metric, predicting a constant downward trend even below 0%.

To solve this issue, we developed a Long Short-Term Memory (LSTM) model to predict future NDVI maps based on historical map data. The model utilizes previous NDVI maps to forecast subsequent maps, capturing temporal dependencies and patterns in the data. Each NDVI map is represented as a 2D array of pixel values, which serves as the input to the LSTM model.

The predicted NDVI Map is then used to calculate the predicted FCL normally.

Model Architecture:

The LSTM model architecture consists of several layers designed to process sequential NDVI data effectively:

- **Input Layer:** This layer receives the input NDVI maps, where each map is represented as a 2D array of pixel values.
- **LSTM Layers:** Multiple LSTM layers are used to capture temporal dependencies in the sequential NDVI data. These layers process the input sequences and learn patterns over time.
- **Dense Layer:** A dense layer with an output size matching the input NDVI map dimensions is used to generate the predicted NDVI map for the next time step.

Loss Function:

The loss function for training the LSTM model is designed to minimize the prediction error for both individual pixels and the overall greenery percentage. The loss function is defined as:

$$\text{Loss} = \sum_{i=1}^N \sum_{j=1}^M |\text{Pred}_{ij} - \text{True}_{ij}| + \alpha \cdot \left(\frac{(\text{FCL}_{\text{Pred}} - \text{FCL}_{\text{True}})^2}{\text{FCL}_{\text{True}} \cdot \text{FCL}_{\text{Pred}}} \right)$$

where:

- Pred_{ij} and True_{ij} are the predicted and true NDVI values at pixel (i, j).
- FCL_{Pred} and FCL_{True} are the predicted and true FCL greenery percentages.
- α is a weight parameter that balances the pixel-wise L1 error and the FCL-based custom loss.

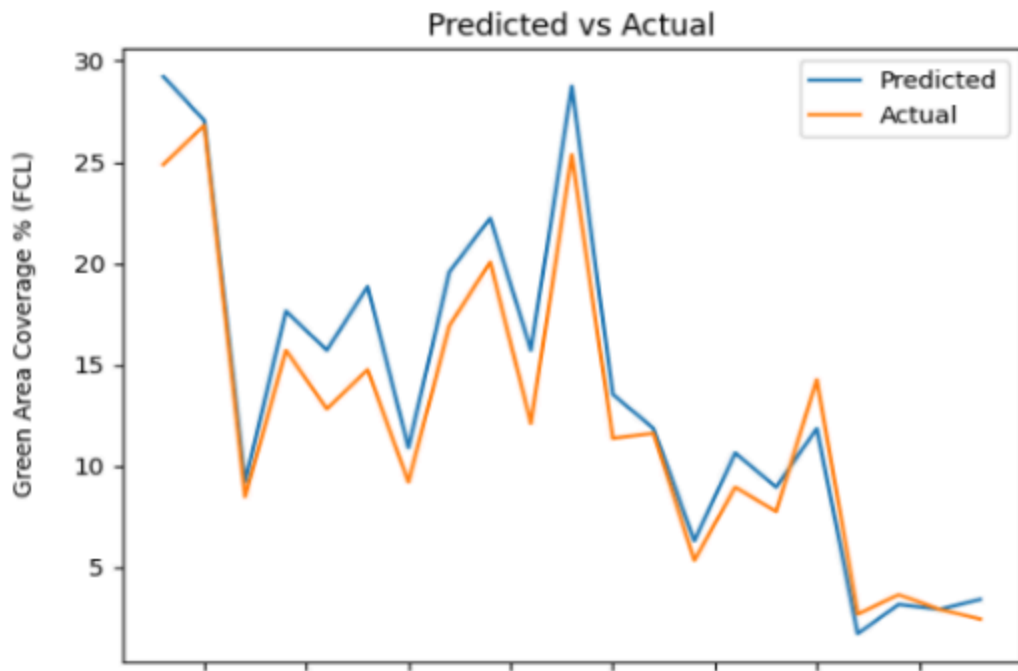


Fig VII.1 Performance on Training Data(1995 to 2018)

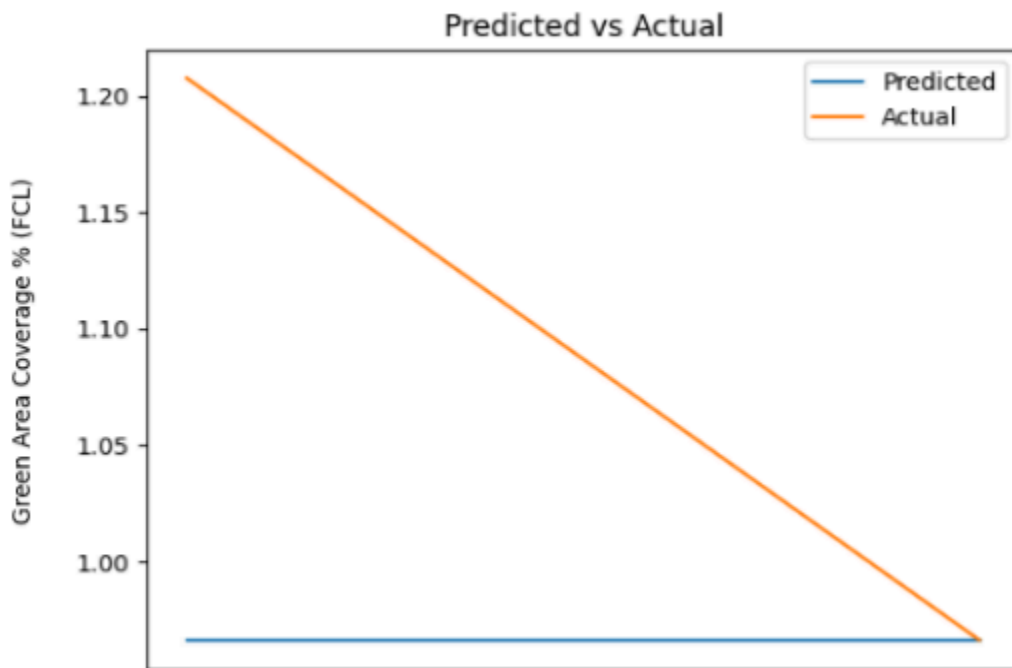


Fig VII.2 Prediction of Test Data (2023 and 2024)

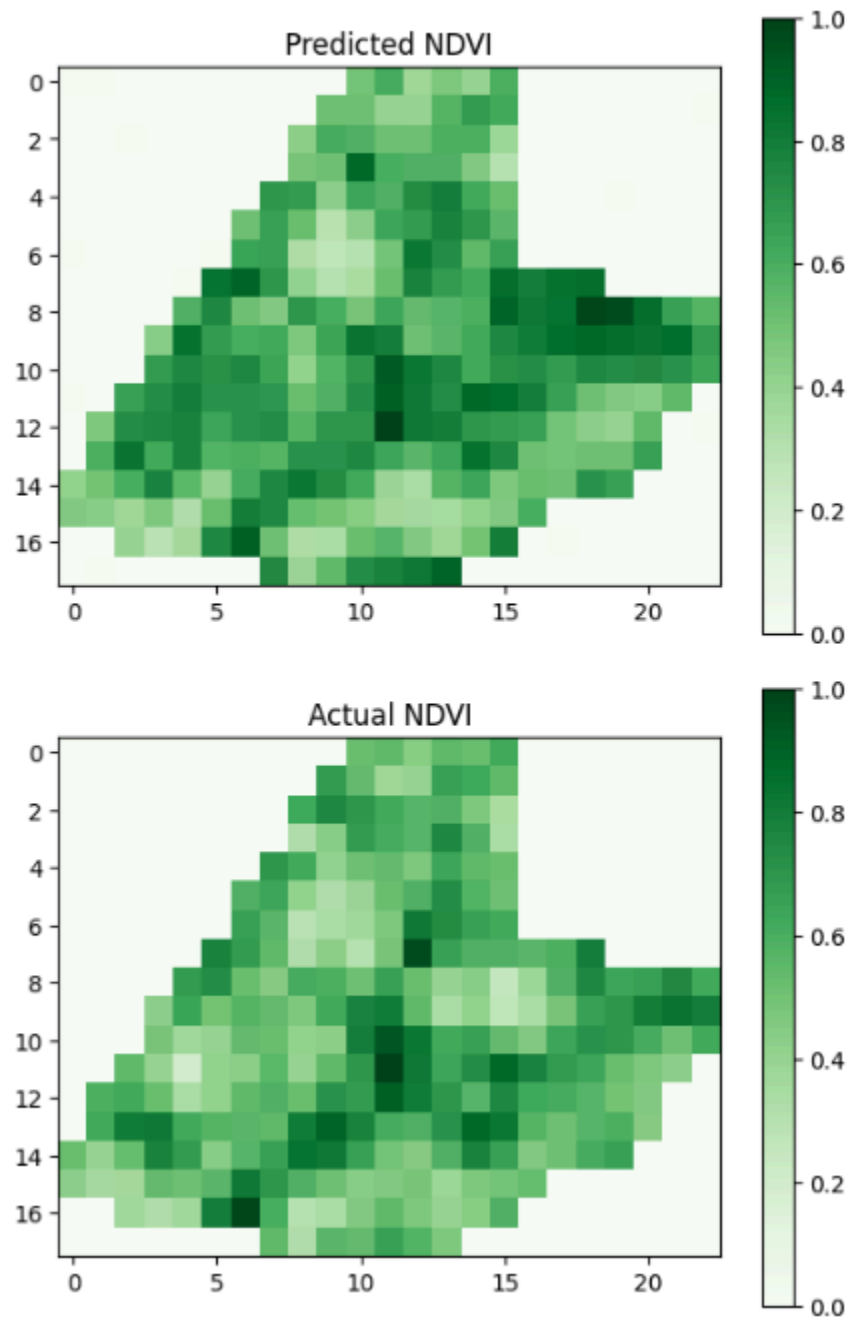


Fig VII.3 Comparison of Predicted and Actual NDVI Maps of 2024

VIII. Conclusion

In this project, we effectively utilized Landsat satellite imagery to analyze and predict the greenery trends on the K.J. Somaiya College of Engineering campus. By employing QGIS for data extraction and preprocessing, and leveraging the LSTM model for future NDVI predictions, we were able to establish a robust framework for monitoring vegetation health over an extended period.

Key Findings:

1. NDVI Trends:

- Analysis of historical NDVI data from 1988 to 2024 revealed significant trends and patterns in vegetation health.
- The introduction of the Fixed Coordinate Limit (FCL) metric provided a novel way to measure consistent greenery over time.
- The Green Area Coverage of the campus has relatively declined from over 39% in 1989 to less than 1% in 2024 with respect to FCL. A major cause of this decline is increase in infrastructure.

2. LSTM Model Performance:

- The LSTM model demonstrated effectiveness in capturing temporal dependencies in the NDVI data, allowing for accurate predictions of future vegetation health.
- The model's architecture, incorporating multiple LSTM layers and a dense output layer, was crucial in generating reliable NDVI forecasts.

3. Environmental Implications:

- The results indicated periods of both improvement and decline in vegetation health, highlighting potential environmental changes or degradation on campus.
- These insights can inform future environmental management and sustainability efforts at K.J. Somaiya College of Engineering.

Challenges:

IX. Data Quality:

Inconsistencies and anomalies in the satellite images, particularly for the years 2020-2021, posed significant challenges. These anomalies were substantial enough that some results were deemed unreliable and excluded from the final dataset.

X. Model Limitations:

While the LSTM model performed well, further enhancements could improve its accuracy and robustness, especially in handling anomalous data points. Sudden changes in greenery due to external factors such as construction of new infrastructure cannot be predicted accurately by the model.

XI. Future Work

Building on the findings of this project, several areas for future work have been identified:

1. Enhanced Data Collection:

The availability of higher-resolution satellite images would enhance the resolution of the NDVI maps and thus improve the accuracy of NDVI calculations and vegetation analysis.

Expanding the dataset to include more recent years and other seasonal images could provide a more comprehensive view of vegetation dynamics.

2. Model Improvement:

Using a more sophisticated LSTM architecture or even a Transformer model might help capture the intricate trends for each pixel in the NDVI map.

Implementing a more sophisticated anomaly detection system to handle outliers and inconsistent data points would improve the model's robustness.

3. Incorporating more metrics affecting vegetation for enhanced prediction:

Regional Metrics like Rainfall measured, Temperature, etc can be combined with NDVI to provide a more accurate estimate of future vegetation coverage.

Investigating the impact of specific environmental factors, such as urban development or climate change, on NDVI trends could provide deeper insights into the causes of vegetation changes.

4. Improvement in FCL metric:

The FCL metric used for estimating the green area coverage has the potential to be further improved or modified to yield more accurate greenery estimation values.

XII. References

- A Machine Learning Approach for NDVI Forecasting Based on Sentinel-2 Data
<https://www.scitepress.org/Papers/2021/105445/105445.pdf>
- GDAL and PROJ Libraries Integrated with GRASS GIS for Terrain Modelling of the Georeferenced Raster Image
<https://www.mdpi.com/2227-7080/11/2/46#:~:text=The%20GRASS%20GIS%20shows%20the.to%20the%20technical%20cartographic%20development.>
- Okinawa Trough Geophysical and Topographic Modeling by GDAL Utilities and GRASS GIS
https://www.researchgate.net/publication/353295404_Okinawa_Trough_Geophysical_and_Topographic_Modeling_by_GDAL_Utilities_and_GRASS_GIS
- Evolution and use of remote sensing in ecological vulnerability assessment: A review
<https://www.sciencedirect.com/science/article/pii/S1470160X23002418>
- A Machine Learning Approach for Mapping Forest Vegetation in Riparian Zones in an Atlantic Biome Environment Using Sentinel-2 Imagery
<https://www.mdpi.com/2072-4292/12/24/4086>
- A GIS-based assessment of green space accessibility: case study of Dundee
https://www.researchgate.net/publication/341678100_A_GIS-based_assessment_of_green_space_accessibility_case_study_of_Dundee
- Satellite Image Processing by Python and R Using Landsat 9 OLI/TIRS and SRTM DEM Data
<https://www.mdpi.com/2313-433X/8/12/317>
- A Sustainable Evaluation Study Based on Emergy–Geographic Information System (GIS) Methodology in Hangzhou City from 2010 to 2035
<https://www.mdpi.com/2075-5309/13/10/2445>
- Application of remote sensing and machine learning algorithms for forest fire mapping in a Mediterranean area
Reference Link: <https://www.sciencedirect.com/science/article/pii/S1470160X21005343>
- NDVI Forecasting Model Based on the Combination of Time Series Decomposition and CNN-LSTM
<https://link.springer.com/article/10.1007/s11269-022-03419-3#:~:text=Results%20indicate%20that%20the%20TSD,%2C%200.9617%20and%200.0447%2C%20respectively.>
- NDVI and a simple model of deciduous forest seasonal dynamics
<https://www.sciencedirect.com/science/article/pii/S0304380001003544>
- Relations between NDVI and tree productivity in the central Great Plains
<https://www.tandfonline.com/doi/abs/10.1080/0143116032000160499>

- An empirical model for prediction of wheat yield, using time-integrated Landsat NDVI:-<https://www.sciencedirect.com/science/article/pii/S0303243418300692>
- A Case Study on a Combination NDVI Forecasting Model Based on the Entropy Weight Method:- <https://link.springer.com/article/10.1007/s11269-017-1692-8>
- Study of mesoscale NDVI prediction models in arid and semiarid regions of China under changing environments:-
<https://www.sciencedirect.com/science/article/pii/S1470160X21008633>
- NDVI Forecasting Model Based on the Combination of Time Series Decomposition and CNN – LSTM:- <https://link.springer.com/article/10.1007/s11269-022-03419-3>
- Prediction of vegetation dynamics using NDVI time series data and LSTM:-
<https://link.springer.com/article/10.1007/s40808-018-0431-3>
- Development of Garlic & Onion Yield Prediction Model on Major Cultivation Regions Considering MODIS NDVI and Meteorological Elements:-
<https://koreascience.kr/article/JAKO201732839399927.page>
- A Forecast Model Applied to Monitor Crops Dynamics Using Vegetation Indices (NDVI):- <https://www.mdpi.com/2076-3417/11/4/1859>
- Development of Korean Paddy Rice Yield Prediction Model (KRPM) using Meteorological Element and MODIS NDVI:-
<https://koreascience.kr/article/JAKO201218559654726.page>
- Crop yield forecasting on the Canadian Prairies using MODIS NDVI data:-
<https://www.sciencedirect.com/science/article/pii/S0168192310003138>
- NDVI as a potential tool for forecasting changes in geographical range of sycamore (*Acer pseudoplatanus* L.): -<https://www.nature.com/articles/s41598-023-46301-x>
- Using earth observation-based dry season NDVI trends for assessment of changes in tree cover in the Sahel:-<https://www.tandfonline.com/doi/abs/10.1080/01431161.2014.883104>
- Forecasting tree mortality using change metrics derived from MODIS satellite data:-
<https://www.sciencedirect.com/science/article/abs/pii/S0378112709003995>